Neil Dewan

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Interests: Experimentation, causal inference, applied ML, big data, data visualization, data analytics
Coursework: Applied Statistics, Econometrics, AI/Machine Learning, Recommender Systems, Natural Language Processing, Data Visualization

Education

University of California San Diego — La Jolla, CA
B.S. in Data Science, Economics Minor — *Expected June 2027*

Hitotsubashi University — Tokyo, Japan
Exchange Program, Social Data Science — *March 2026 – July 2026*

Skills

Languages: English (Native), Japanese (Native) — U.S. and Japan citizenship
Tech Stack: Python (Pandas, scikit-learn, NumPy), SQL (PostgreSQL), R, Stata, PySpark, JavaScript, HTML, D3.js
Statistics: Hypothesis testing (ANOVA, permutation tests), regression, econometrics, k-means clustering
AI / Cloud: Claude Code, Gemini (Vertex AI), LangGraph, Elasticsearch, TensorFlow, Google Cloud Run, AWS EC2

Work Experience

AVILEN Inc. — Tokyo, Japan
Data Science Intern — *Apr 2026 – Oct 2026*

- Built the prompt-engineering, feature engineering, and evaluation layer of an automated computer vision PoC extracting instrument data from engineering blueprints (P&ID) for an enterprise client; raised accuracy on a 238-item evaluation set from **80.7% to 93.3%** through legend-derived prompt rules, crop-geometry tuning, and a reference image.
- Performed root-cause error analysis on the pipeline's extraction failures, isolating data-quality issues from pipeline issues.
- Led the client-facing side of a two-person team pitching a workflow automation tool prototype to a Japanese electronics manufacturer and AVILEN leadership; in timed trials it reduced the client's stated **1-hour task to 5 minutes**.

UC San Diego Jacobs School of Engineering — La Jolla, CA
Corporate Relations Assistant — *September 2025 – Present*

- Managed corporate job postings on LinkedIn and campus job portals to reach student candidates, tracked posting analytics, and designed promotional materials in Canva and Adobe Acrobat.
- Coordinated and executed **50+** in-person events across **20+** corporate affiliates, working directly with recruiters, executives, and engineers; served as the on-site point of contact for check-in, guest support, and day-of coordination.

Projects

UCSD Parking Occupancy Analysis (DSC 190 Statistical Inference Project)

- Analyzed 20 years of **1,071** peak-hour occupancy observations across all UCSD parking spaces and 5 permit types with one-way ANOVA and a 3,000-shuffle permutation test ($\boldsymbol{F = 25.52}$, $\boldsymbol{p < 0.001}$).
- Coordinated with UCSD Transportation Services President and presented findings such as resident-student parking near saturation (**97.5%** mean) vs. **87.6%** for commuter students.

Scalable Feature Engineering & Model Tuning (DSC 102 Course Project)

- Built an Apache Spark pipeline on a 3-worker Kubernetes cluster over a **120GB** Amazon product and review dataset.
- Flattened nested schemas, conducted EDA, trained Word2Vec title embeddings, applied one-hot encoding, and tuned a decision tree regressor to predict user ratings, emphasizing scalable and parallel computation.

Multilingual News Divergence Agent (Google Cloud Rapid Agent Hackathon) Devpost, **14,000+** participants

- Built an agentic pipeline (Gemini 2.5 Flash on Vertex AI, LangGraph, Elasticsearch) that aligns claims across English, French, Greek, and Japanese news articles and surfaces factual discrepancies for humanitarian analysts.
- Cut repeat-query time from **∼90s to ∼30s** via article dedup, index reuse, and parallel fetching; surfaced conflicting casualty counts across named sources on events such as the 2023 Libya flood case and 2024 Noto earthquakes in Japan.

Ukraine Crisis Map (International Humanitarian Mapathon) — **Best Design, 2nd** of 50+ teams

- Co-built (team of 4) an interactive web map in MapLibre GL JS with Claude Code, plotting ACLED missile, drone strike, and IOM displacement data with toggleable humanitarian layers.
- Iterated on the map using feedback from a current UN staff member who works on peacekeeping operations.

Stress and Academic Performance Analysis (Interactive Visualization Project)

- Built and presented an interactive web application visualizing the relationship between physiological stress and exam performance using wearable sensor data; emphasized data storytelling and how students have different stress archetypes.
- Preprocessed the data in Python (Pandas), applied k-means clustering, and designed dynamic visualizations with D3.js.